In this document we wrote the reasoning behind our Simulator\MissionControl\MappingAlgorthim\Common structure. This should help you in assignment 3. Try to understand our way of thought about this design, it might be helpful in other future design tasks.

Simulator

Note that some implementations of the Common folder are stored in the `src` folder of the Simulator. This is not by mistake - the mocks we make are completely simulated, and would be replaced by real API's of real components in real life.

The simulation includes, in its common folder the interfaces for:

- The simulation Manager
- The simulation Run
- The simulation run factory
- Simulation types These are pretty straight forward - they all take part in the "running behind the scenes" aspect of the program. They are not a part of what would have been in real life. In the same spirit, the src folder includes:
- Mock Lidar
- MockGPS
- MockMovement
- Map3DImpl
- Implementations for simulation and code specific objects Because these are a part of the simulation, their implementation can be unrealistic. For example, from the first assignment we know that MockLiDAR needs to hold the real, hidden map. We can give that map to both MockGPS and MockMovement if they really need it (and only one of them might!).

Mission Control

The common includes only the IDroneControl. This is because according to assignment 3, it is the responsibility of the Mission Control to bring its own implementation. Note that both could (with minimal changes) be implemented in a micro controller of some real life device. The actual interface of MissionControl is stored in the Common folder! Because it is used by the simulation, too.

Mapping algorithm

Similar to the MissionControl, There is no interface in the common folder of Mapping algorithm, which is actually stored in the Common folder. Again because other objects need that interface, too.